

Ka-Boom!!! Visually Exploring Latency Measurements for XR

Jan-Philipp Stauffert
University of Würzburg

Florian Niebling
University of Würzburg

Kristof Korwisi
University of Würzburg

Marc Erich Latoschik
University of Würzburg

ABSTRACT

Latency can be detrimental for the experience of Virtual Reality. High latency can lead to loss of performance and cybersickness. There are simple approaches to measure approximate latency and more elaborated for more insight into latency behavior. Yet there are still researchers who do not measure the latency of the system they are using to conduct VR experiments.

This paper provides an illustrated overview of different approaches to measure latency of VR applications, as well as a small decision-making guide to assist in the choice of the measurement method. The visual style offers a more approachable way to understand how to measure latency.

CCS CONCEPTS

• General and reference → Measurement; • Computing methodologies → Virtual reality.

KEYWORDS

VR, XR, latency, measurement, comic

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Latency can provoke **unpleasant effects** and **dangerous situations**.



If latency is too high, people lose performance [9]



or get sick [17].

They may hit a wall they thought was still far away.

Or worse, injure uninvolved third parties [23].



To effectively combat latency, we must first find it, i.e. measure it. Only then we can take appropriate action.

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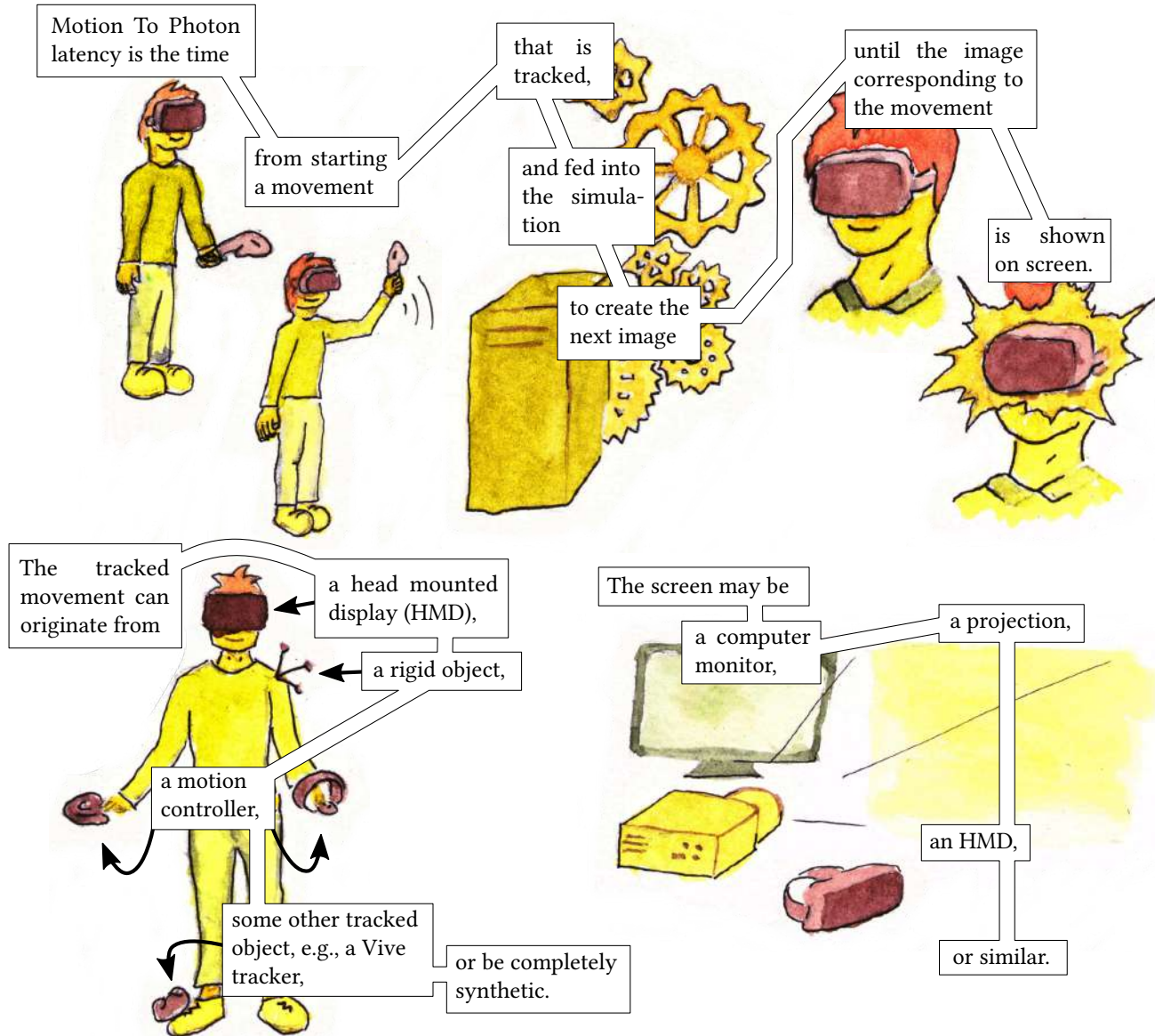
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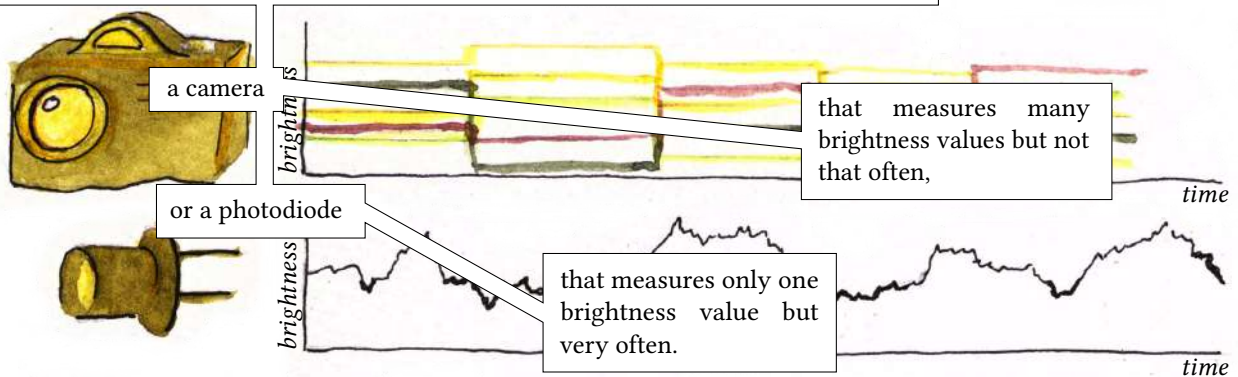
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When talking about latency, we usually refer to **Motion To Photon Latency** or End To End Latency.



For measurement, both the movement and its effect on screen are captured by either



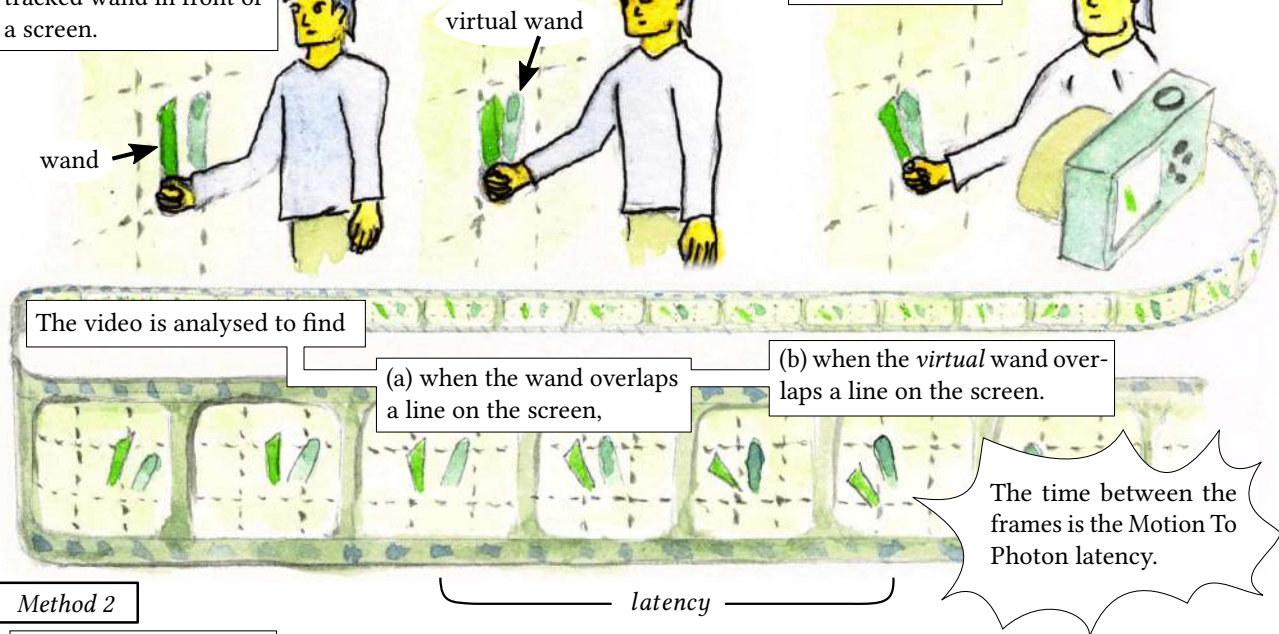
The most straightforward form of latency measurement is **Frame Counting**.

Method 1

He et al. [8] move a tracked wand in front of a screen.

The screen shows the virtual counterpart.

A camera records the scene.



Method 2

Similar, Wu et al. [24] use a moving bar.

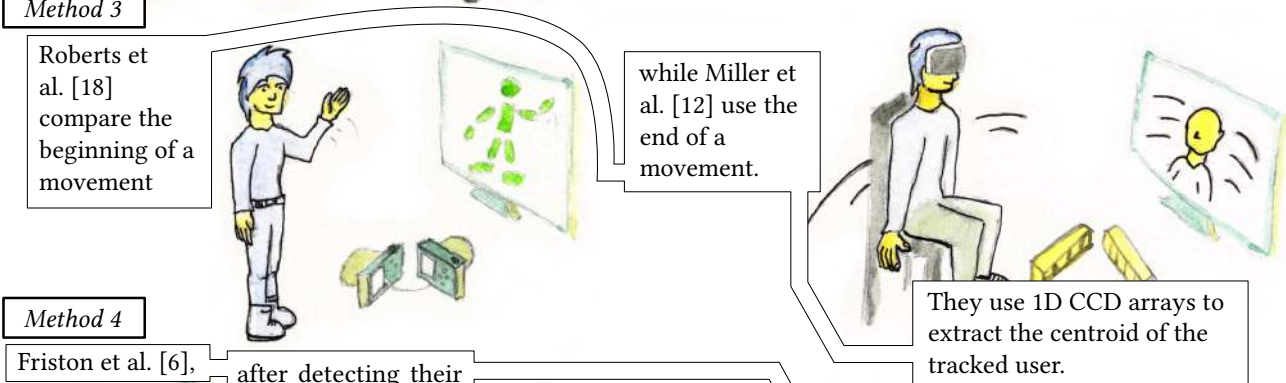
The position of the real and virtual bar can be extracted automatically via thresholding.



Method 3

Roberts et al. [18] compare the beginning of a movement

while Miller et al. [12] use the end of a movement.

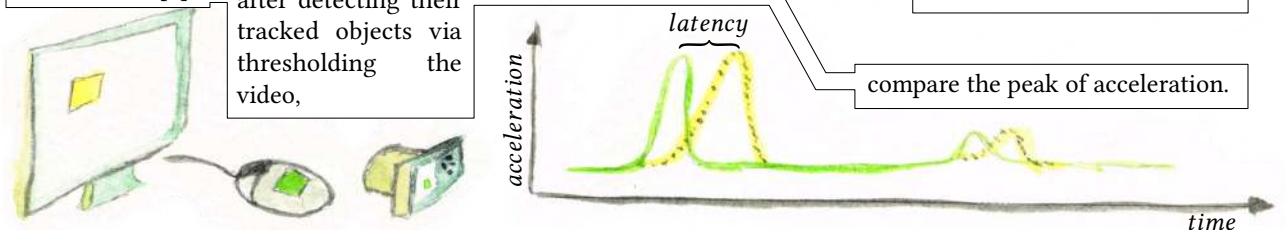


Method 4

Friston et al. [6],

after detecting their tracked objects via thresholding the video,

compare the peak of acceleration.



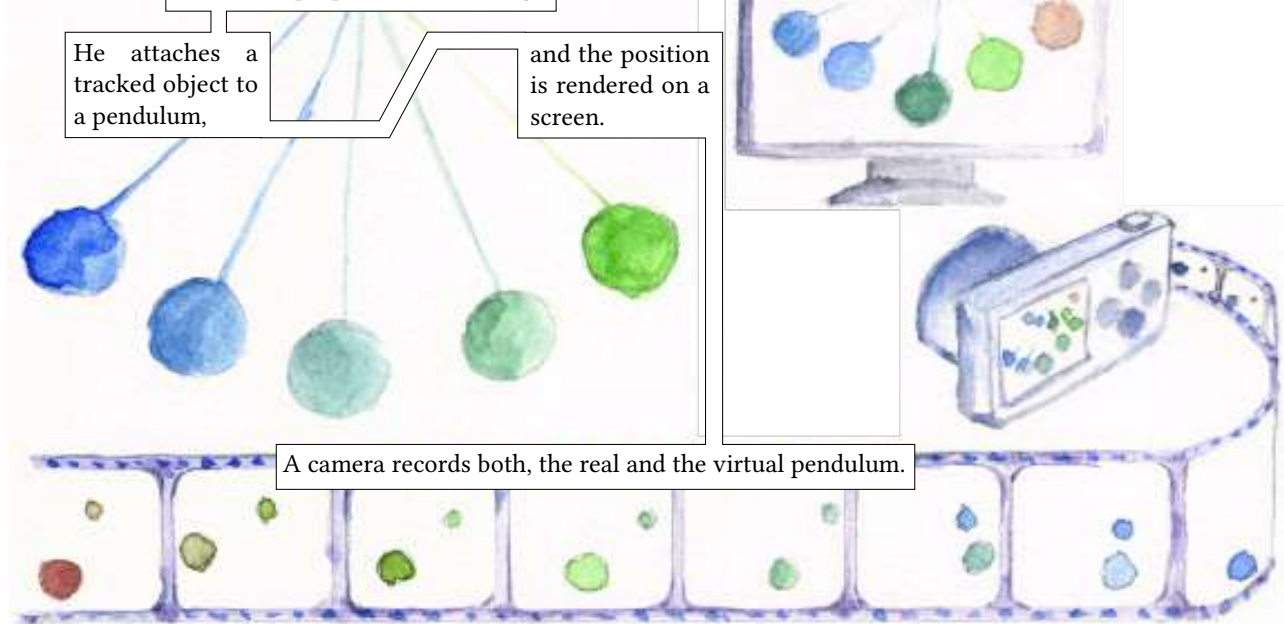
Many approaches use a **Pendulum**

Method 1

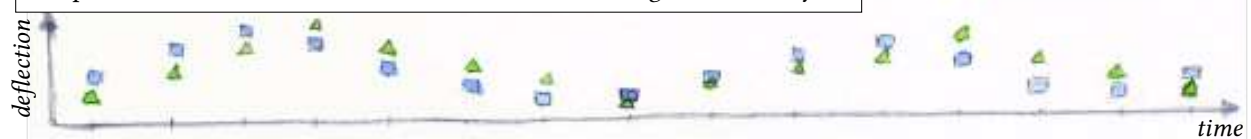
Steed [21] proposes **Sine Fitting**:

He attaches a tracked object to a pendulum,

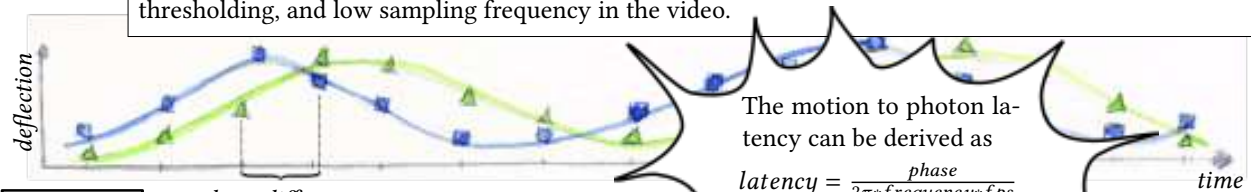
and the position is rendered on a screen.



The positions are extracted from the video via thresholding automatically:



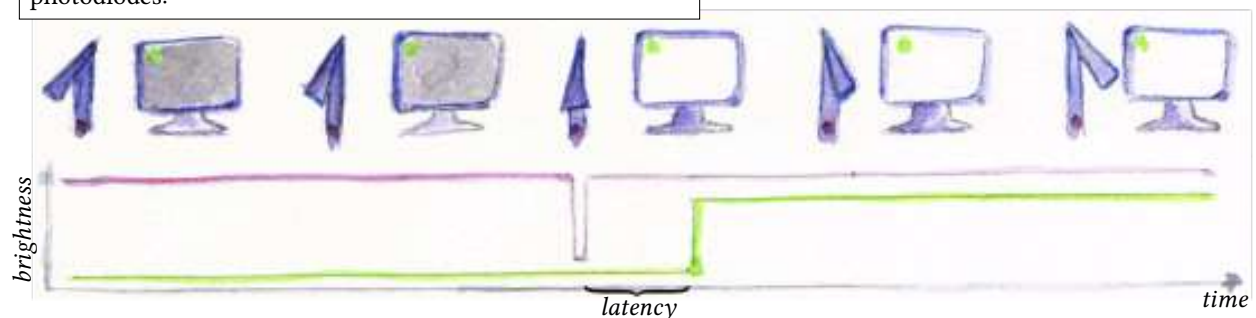
He fits a sine curve (e.g. $\cos(2\pi * \text{frequency} * \text{deflection} - \text{phase})$). The fit prevents inaccuracies of the thresholding, and low sampling frequency in the video.



Method 2

phase difference

The predictability of a pendulum is used in other approaches to measure latency. Mine et.al. [13] uses a pendulum and two photodiodes:



Other approaches

Method 1

Liang et al. [11] record a pendulum and the timestamp of the last tracking data.

Once the pendulum overlaps the reference

they look how much the tracking data deviated at the recorded timestamp.

Method 2

Chang et al. [3] rotate an HMD.

The virtual scene mirrored on a monitor follows with a delay to the rotation.

Bars drawn on the HMD make it easier to recognise motion.

The VR system changes the brightness of a projected dot once it receives the motion information.

Method 3

Pape et al. [15] use projection based VR.

The ground truth when the servo motor starts to move is provided with a switch.

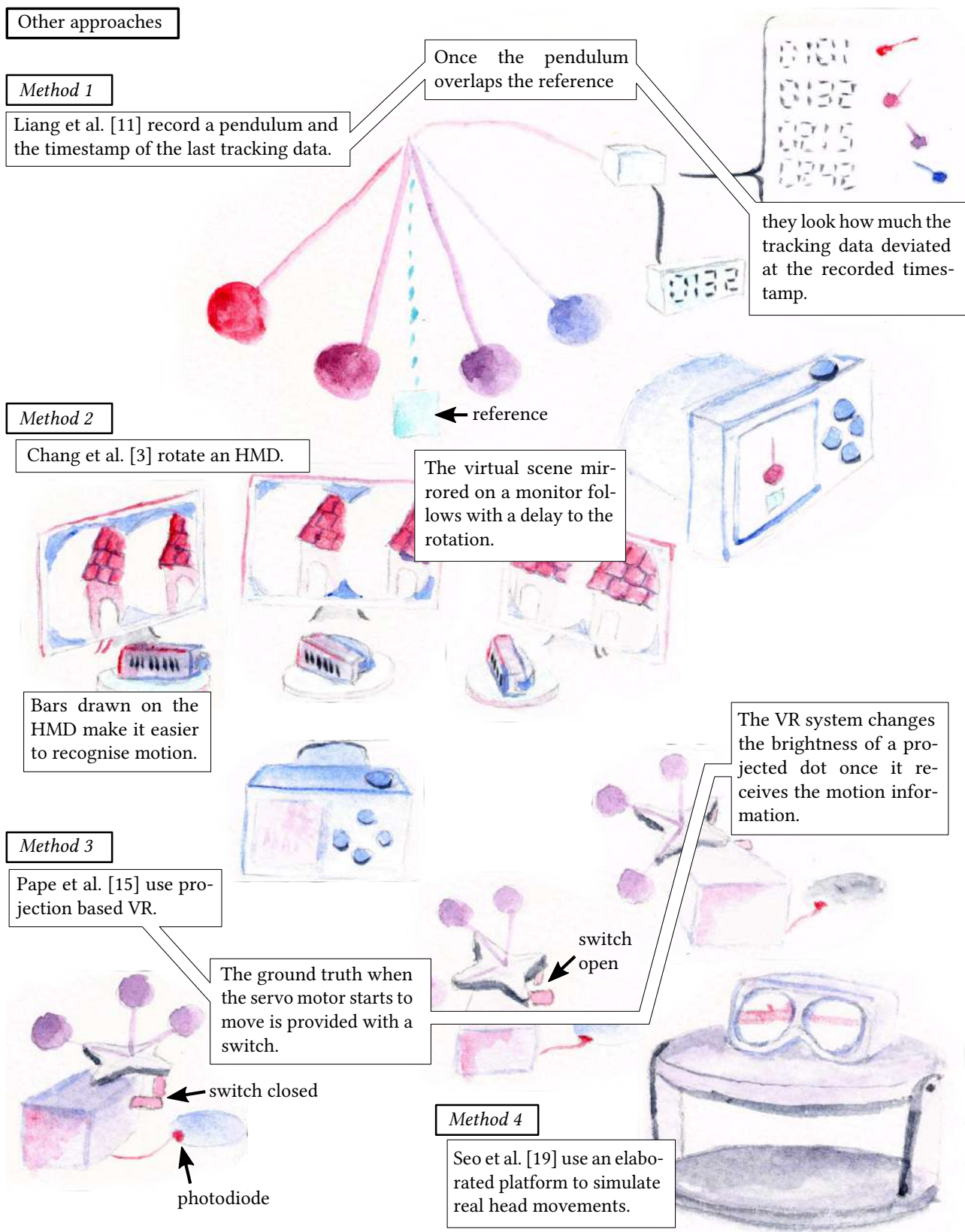
switch open

switch closed

photodiode

Method 4

Seo et al. [19] use an elaborated platform to simulate real head movements.



Measuring latency with HMDs is difficult because

the image of the screen

is distorted by lenses

which creates a scrambled image if the camera is not right in front of the lenses.

Method 1

Feldstein et al. [5]

pop out the lenses

then push the HMD.

The scene follows the HMD movement with a delay.

plop

push

Method 2

Kijima et al. [10] use two synchronised cameras.

One observes a real stick

Upon rotation, the real one moves faster out of the image

while the other records the virtual counterpart.

than the virtual counterpart

which is used to determine the latency.

Method 3

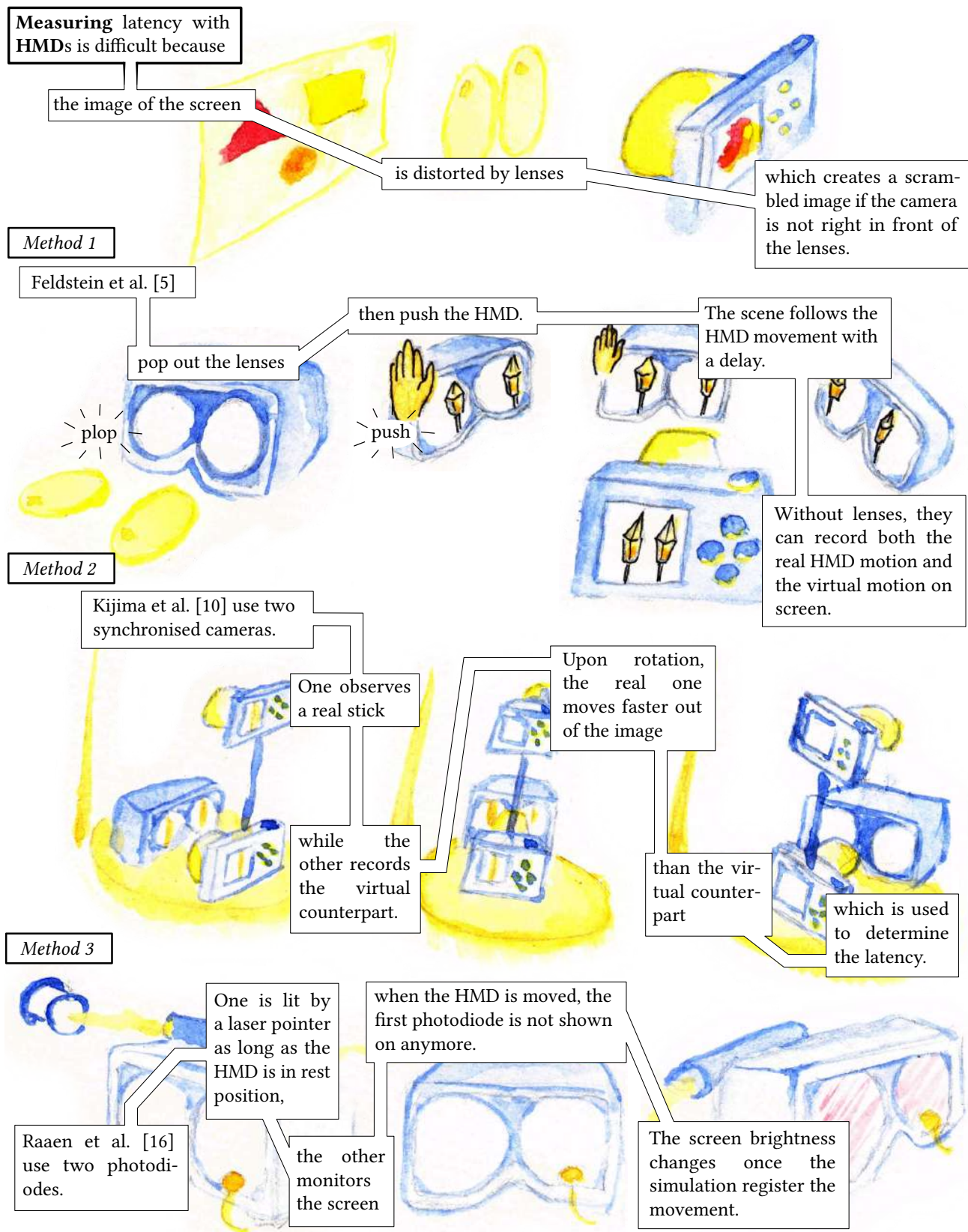
Raaen et al. [16] use two photodiodes.

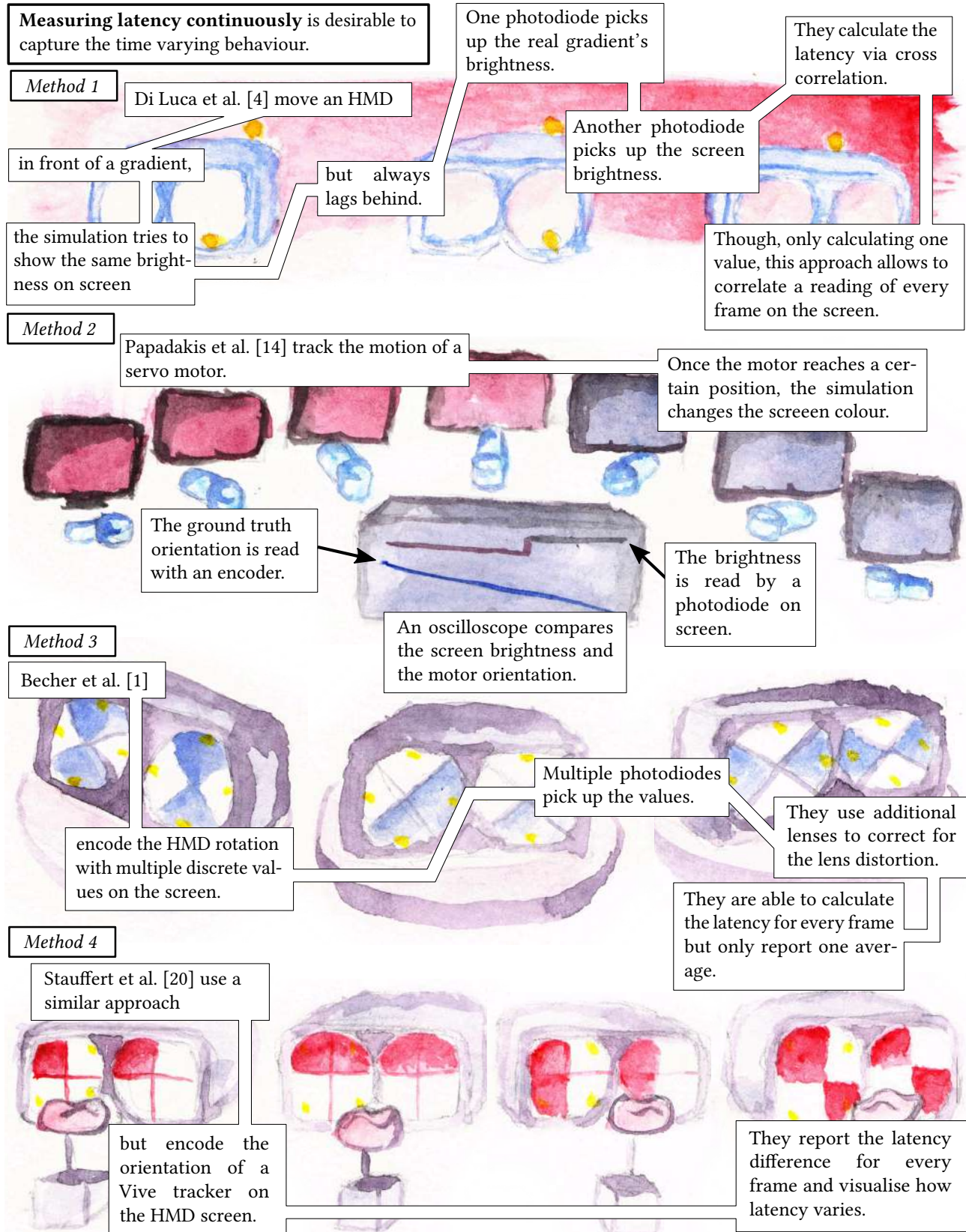
One is lit by a laser pointer as long as the HMD is in rest position,

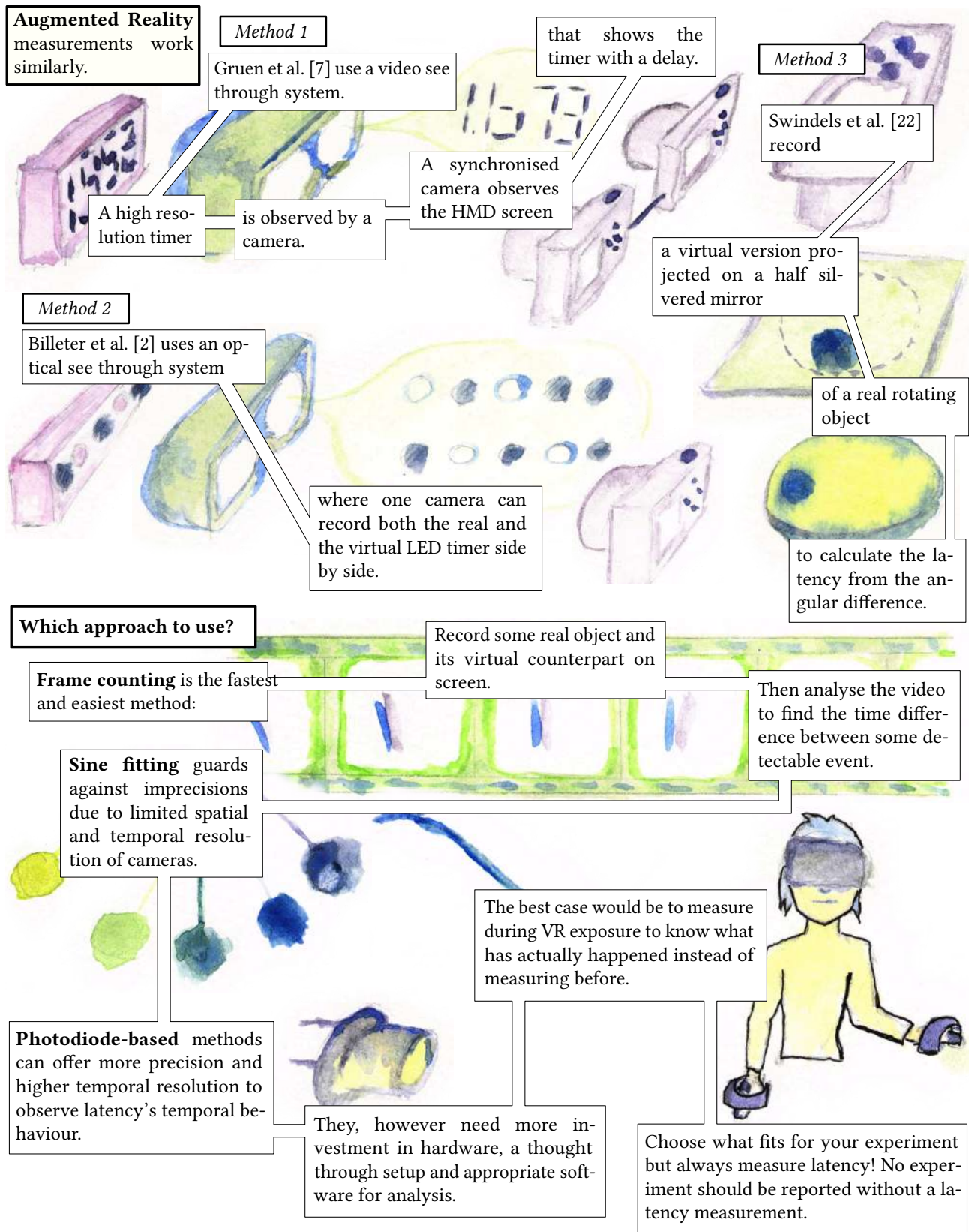
when the HMD is moved, the first photodiode is not shown anymore.

the other monitors the screen

The screen brightness changes once the simulation register the movement.







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